

Anxiety and Affective Symptoms Related to the Use of Classic Psychedelics: A Systematic Review



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Abstract There is a large and rapidly growing body of literature investigating the therapeutic effects of classic psychedelics in affective and anxiety disorders, but very few studies have examined the inverse of this, that is, the potential for psychedelics to inflict anxiety and affective symptoms. A systematic literature search was performed and 39 papers were included in the final review to qualitatively synthesize

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the current literature on anxiety and affective disorders related to the use of classic psychedelics. Persisting disorders were less frequent but generally occurred in individuals who presented with several risk factors (overdose, polydrug use, unstructured recreational setting, psychosocial stress, personal/familial psychiatric histories). When psychedelics were administered in clinical studies under the framework of psychedelic-assisted therapy, the incidence of enduring anxiety and affective symptoms was low. In most cases, acute transient anxiety emerged and resolved during the dosing session without the need for additional treatment interventions. The nuance of such cases is discussed, shedding light on the role of emotional catharsis in the therapeutic process. Several suggestions are proposed to enhance patient safety including strengthening the therapeutic alliance, ensuring adequate mental preparation, acclimating to high doses and providing on-going therapeutic support.

Keywords Affective disorders · Anxiety · Depression · LSD · Psilocybin · Psychedelics

1 Introduction

Lysergic acid (LSD), psilocybin, dimethyltryptamine (DMT) and mescaline share a common ability to elicit marked changes in sensory-perceptual experiences. These drugs are termed classic psychedelics and produce similar subjective effects yet vary widely depending on the dose. For example, at a low dose, psychedelics have been shown to evoke subtle intensification in sensory perception or at high doses, they may completely dissolve all sense of time, space and self (Swanson 2018). Aside from dosage, set and setting may account for this diversity of subjective effects (e.g. the mood of the individual and the environment in which the drug is taken). The prevalence of psychedelic use is on the rise, with findings from the European Web Survey on Drugs showing that 20% of respondents engaged in LSD use in the last year ($n = 48,469$) (EMCDDA 2021).

It is not only the high and rising prevalence which makes the topic relevant; the psychedelic renaissance over the last decade has made us well aware of its therapeutic effects and medicinal applications. There is a large and rapidly growing body of literature investigating therapeutic aspects of psychedelics in the treatment of psychiatric disorders (Gasser et al. 2014; Bogenschutz et al. 2022; Goodwin et al. 2022). To date, the clinical efficacy of psychedelic therapy has been established in major depressive disorder and anxiety related to life-threatening cancer diagnoses with research continuing in other disorders such as addiction, obsessive-compulsive disorder and anorexia nervosa (Lowe et al. 2021). With several late-stage clinical trials underway it is suggested that psychedelic-assisted therapy may be a treatment option available to those suffering from psychiatric disorders in a matter of years (Mertens et al. 2022). However, if psychedelic-assisted therapy is to be seriously

considered, not only in the realms of psychiatry but also as a public health matter, it is important to explore the potential for psychedelics to cause harm. This becomes increasingly important when considering that psychedelics are not exclusively used within the confines of treatment contexts. Thus, the potential adverse effects of psychedelics should be studied directly to characterize perceptions of risk and danger accurately.

It is striking that scientific attention devoted to understanding the potential for psychedelics to cause harm has been very scarce. There is a significant shortage of papers which address the harmful potential of psychedelics in the context of anxiety and affective disorders. However, although limited, the literature suggests that such a phenomenon may exist. One of the few studies assessing this beyond a case-by-case basis was a correlational study comprising 4,745 people. Bonn-Miller et al. (2007) examined the association between different levels of psychedelic use and a lifetime history of panic attacks. The study found that psychedelic abuse and dependence, but not use, were significantly related to an increased lifetime risk of panic attacks.

More formative studies have shown that psychedelics may also increase symptoms of anxiety and depression. A recent study assessed psychedelic use in relation to psychological distress, depression and suicidality and found that LSD use was associated with an increased likelihood of a major depressive episode (MDE) and suicidal thinking whilst N-N-DMT use was associated with an increased likelihood of suicidal planning (Yang et al. 2022). In randomized controlled trials (RCTs) that have used psychedelics, high doses have been shown to significantly increase transient anxiety. Studerus et al. (2011) pooled data from eight RCTs to assess the acute, subacute and long-term subjective effects of psilocybin in healthy participants. The study showed that acute symptoms of anxiety and panic manifested in 5/110 of the patients that were in the highest dose condition (315 µg/kg), relative to low-medium doses (45–260 µg/kg). Another study in healthy subjects found that the 200 µg LSD induced significant transient anxiety (Holze et al. 2021). Collectively, these studies highlight the capacity of psychedelics to trigger symptoms of depression and anxiety under the acute effects of the substance, particularly at high doses, as well as panic attacks and depressive symptoms.

These findings necessitate a further investigation of psychedelic-related anxiety and affective disorders. Thus, we conducted a systematic review which aimed to summarize the current literature on this phenomenon. Despite current trials also investigating the therapeutic potential of atypical hallucinogens (e.g. ketamine) or entactogens (e.g. MDMA), the scope of this review was limited to classic psychedelics which target 5-HT_{2A} receptors as the sole neurochemical mechanism (e.g. psilocybin, LSD, DMT). In synthesizing the existing literature, we were interested in identifying the possible risk factors for psychedelic-related symptoms. This systematic review is the first to comprehensively summarize the available literature on the association between psychedelic drugs and subsequent anxiety and affective symptoms.

2 Methods

2.1 Search Strategy

This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines (PRISMA) (Page et al. 2021). The protocol of this study is registered on PROSPERO (record number: CRD42024512599). From February to March 2024, we performed a systematic literature search using the PubMed, Cochrane and Scopus databases. The search was conducted using keywords related to classic psychedelics (e.g. psilocybin, LSD, DMT, ayahuasca, mescaline, peyote) and anxiety and affective disorders (e.g. anxiety, major depressive disorder (MDD), bipolar disorder (BP), mania, obsessive-compulsive disorder (OCD), post-traumatic stress disorder (PTSD), panic disorder, phobia, dysthymia, cyclothymic disorder, premenstrual dysphoric disorder, substance-induced mood disorder). Eligibility criteria included the use of classic psychedelic drugs and the emergence of anxiety and/or affective disorder symptoms during or after the psychedelic experience. A filter was applied to in-human studies only and no date restrictions were applied. The search was limited to studies published in English in peer-reviewed journals. All study methodologies were considered, ranging from case reports to RCTs. Articles examining the use of microdosing, however, were excluded from the review since such doses elicit responses below the perceptual threshold, rendering it too difficult to establish a definitive association (Bershad et al. 2019). Study selection was performed by G.V. and verified by F.B.

2.2 Data Extraction and Risk of Bias Assessment

For studies that met inclusion criteria, data regarding the specific classic psychedelic drug used and related disorders/symptoms following drug use were extracted. If applicable, various patient characteristics (age, sex, personal and familial psychiatric history) along with dose and time of symptom onset were also extracted. Finally, we noted the rescue medication used to treat subsequent anxiety and affective symptoms in studies that stated it. To evaluate multiple study designs, risk of bias was performed using an array of validated quality assessment tools. Non-randomized trials were assessed using the modified Newcastle-Ottawa Scale (Deeks et al. 2003) whereas randomized trials were assessed using the Cochrane Risk of Bias tool (Sterne et al. 2019). For case reports and observational studies, the Joanna Briggs Institute checklist for case reports (Moola et al. 2020) and the CASP cohort studies checklist (CASP 2018) were employed, respectively. Quality assessment was performed by G.V. and verified by F.B.

3 Results

3.1 Article Identification Process

The systematic search in March 2024 identified 1,034 unique articles which were screened based on titles and abstracts. Nine hundred and nineteen papers were excluded after initial screening and the full text of 115 articles was assessed for eligibility. Following the exclusion of 41 articles that lacked reports of harm, 24 articles classified as reviews or meta-analyses and 11 studies that involved microdoses, the final review included 39 papers (Fig. 1). The year of publication for the included articles ranged from 1969 to 2024. A quantitative literature analysis was deemed inappropriate given the heterogeneity in the study methodology. A large proportion of the literature on this topic consisted of clinical trials (59%) with the rest comprising qualitative case reports (28%), surveys (8%) and cross-sectional or observational studies (5%). Thus, it seemed most suitable to synthesize the extracted data narratively. Data from the 39 papers are presented in text and summarized in

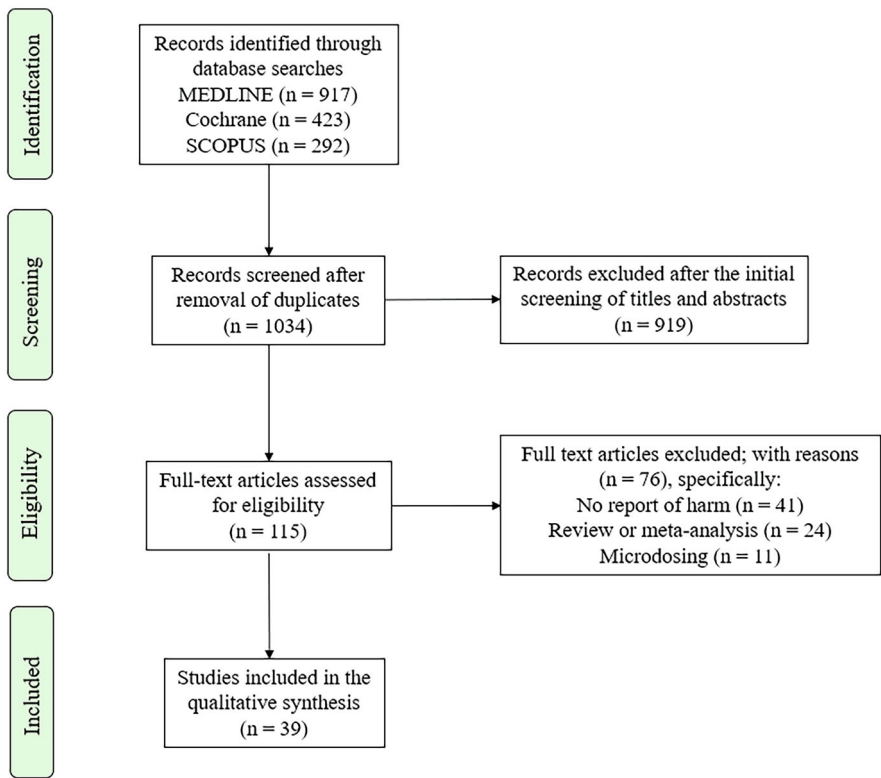


Fig. 1 PRISMA flowchart

table form grouped by psychedelic drug: Psilocybin ($n = 17$), LSD ($n = 9$), DMT ($n = 8$), a combination of classic psychedelics ($n = 5$).

3.2 *Tabulation by Psychedelic Drug*

3.2.1 **Psilocybin**

The search identified 17 studies showing psilocybin-related symptoms of anxiety and affective disorders (Table 1). Three case reports identified psilocybin-related mania warranting the use of antipsychotic medication. Hendin and Penn (2021) reported the case of a 21-year-old woman with a psychiatric history of MDD and PTSD and a family history of BP. After ingesting psilocybin, the patient was hospitalized for mania and was later diagnosed with BP and experienced successful treatment with lamotrigine. Eight months prior to ingesting psilocybin, she was hospitalized for depression and suicidal ideation. Suleiman et al. (2022) reported the onset of mania symptoms following high-dose psilocybin in an 18-year-old male. The patient reported daily marijuana use for 2 years before the psychedelic experience and had a psychiatric history of attention-deficit hyperactivity disorder (ADHD). Another case report documented symptoms of mania and psychosis, followed by MDD after the consumption of psilocybin 2 days in a row (Barber et al. 2022). The 32-year-old female had a personal psychiatric history of MDD and generalized anxiety disorder (GAD) and a family history of bipolar disorder. Panic disorder following psilocybin use was also reported in a case study of a 24-year-old male (Benjamin 1979). It was noted that the patient was experiencing stress in interpersonal relationships and had a history of cannabis use. Symptoms resolved with lorazepam administration (Benjamin 1979). In most of the psilocybin clinical trials included in the review, adverse effects consisted of transient anxiety symptoms (Tables 1, 2, and 3) (Griffiths et al. 2006, 2011, 2016; Hasler et al. 2004; Studerus et al. 2011; Carhart-Harris et al. 2016; Davis et al. 2021; Goodwin et al. 2022). Only one case warranted rescue medication whereby lorazepam was used (Goodwin et al. 2022). Furthermore, three clinical trials in patients with MDD or treatment-resistant depression (TRD) reported depressive symptoms following psilocybin treatment in a small percentage of patients (Goodwin et al. 2022, 2023; Sloshower et al. 2023). Similarly, affective symptoms were also reported in a low number of subjects following psilocybin trials in healthy participants (Rucker et al. 2022) and patients with body-dysmorphic disorder (BDD) (Schneier et al. 2023), neither of which warranted the use of rescue medication. Finally, a prospective longitudinal survey found that 3% of respondents reported depressive symptoms 2–4 weeks following psilocybin use (Nayak et al. 2023). Approximately 29% had a pre-existing mood disorder.

Table 1 Summary of studies examining psilocybin-related symptoms of anxiety and affective disorders

Study	Major findings	Study design	N	Disorder	Treatment
Benjamin (1979)	A 24yo male presented with a three-month history of panic attacks which began 2 weeks after psilocybin overdose	Case report	1	Panic disorder	Lorazepam
Hasler et al. (2004)	Among healthy participants, one subject reacted with transient anxiety following 315 µg/kg psilocybin	Phase I clinical trial	8	Transient anxiety symptoms	n/a
Griffiths et al. (2006)	Among healthy participants, 11 reported strong or extreme ‘experience of fear’ following 30 mg/70 kg psilocybin	Phase I clinical trial	36	Transient anxiety symptoms	n/a
Griffiths et al. (2011)	39% of healthy volunteers reported extreme anxiety/fear during the dosing session (20–30 mg/70 kg psilocybin)	Phase I clinical trial	18	Transient anxiety symptoms	n/a
Studerus et al. (2011)	In healthy participants, dysphoria or anxiety/panic occurred in 5 out of 110 following 250–315 µg/kg psilocybin	Pooled data from 8 RCTs	110	Transient anxiety symptoms	n/a
Carhart-Harris et al. (2016)	In a study of patients with TRD, all participants reported transient anxiety during the drug onset (25 mg psilocybin)	Open-label trial	12	Transient anxiety symptoms	n/a
Griffiths et al. (2016)	In a trial of patients with life-threatening cancer and symptoms of anxiety and/or depression, anxiety was reported by 26% of patients in the high-dose group (30 mg/70 kg). One participant had a transient episode of paranoid ideation	Phase II clinical trial	51	Transient anxiety symptoms	n/a
Hendin and Penn (2021)	A 21yo female with a history of MDD and PTSD was hospitalized for mania after high-dose psilocybin ingestion	Case report	1	Mania	Lamotrigine
Suleiman et al. (2022)	An 18yo male developed symptoms of psychosis and mania following the ingestion of 2 g of psilocybin	Case report	1	Mania	Olanzapine, divalproex sodium and lithium
Davis et al. (2021)	In patients with MDD, transient symptoms of anxiousness and panic were reported in 56% and 40% of participants, respectively	Phase II clinical trial	27	Transient anxiety symptoms	n/a
Rucker et al. (2022)	In healthy participants, 7% reported depressed mood and anxiety following psilocybin administration (10 or 25 mg)	Phase I clinical trial	89	Transient affective symptoms	n/a

(continued)

Table 1 (continued)

Study	Major findings	Study design	N	Disorder	Treatment
Barber et al. (2022)	A 32yo female presented with mania and psychosis which later developed into severe depression following the consumption of psilocybin 2 days in a row	Case report	1	Mania, psychosis and MDD	Pramipexole
Goodwin et al. (2022)	Among patients with TRD, one participant displayed acute anxiety following psilocybin (25 mg). From day 2 to week 3, suicidal ideation was reported by four participants and intentional self-injury by three participants	Phase II trial	233	Anxiety, depressive symptoms	Lorazepam
Schneier et al. (2023)	In a cohort of participants with BDD, one patient reported episodes of emotional and mood disturbances after psilocybin (25 mg)	Phase I trial	12	Mood disturbances	n/a
Slushower et al. (2023)	In a cohort of individuals with MDD, one patient sought psychiatric hospitalization after psilocybin (0.3 mg/kg)	Phase II trial	19	Depressive symptoms	n/a
Goodwin et al. (2023)	Psilocybin (25 mg) was administered for TRD in patients taking a concomitant SSRI medication. From day 2 to week 3, one patient in the cohort reported depressive symptoms	Phase II trial	19	Depressive symptoms	n/a
Nayak et al. (2023)	A prospective survey collected data on adults planning to take psilocybin. Approximately 3% reported depressive symptoms 2-4 weeks after psilocybin	Prospective longitudinal survey	1,182	Depressive symptoms	n/a

Abbreviations: BDD: Body-dysmorphic disorder; MDD: Major depressive disorder; PTSD: Post-traumatic stress disorder; RCT: Randomized controlled trial; SSRI: Serotonin reuptake inhibitor; TRD: Treatment-resistant depression

Table 2 Summary of studies examining LSD-related symptoms of anxiety and affective disorders

Study	Major findings	Study design	N	Disorder	Treatment
Boer and Sippelle (1969)	A college student experienced symptoms of anxiety and panic following LSD ingestion. The author suggested that the symptoms were a displaced manifestation of anxiety triggered by anxiety over separation from family	Case report	1	Anxiety	Psychotherapy
Lake et al. (1981)	A 31 yo man suffered a manic episode 2 weeks following LSD ingestion	Case report	1	Mania	Lithium
Blaho et al. (1997)	All patients (aged 19–27) were admitted for LSD overdose after attending a concert. Adverse reactions included panic reactions and acute anxiety	Case reports	5	Anxiety symptoms	Lorazepam
Lerner et al. (2002)	A 26yo male developed HPPD and a major depressive episode following a history of cannabis, MDMA and LSD use	Case report	1	HPPD and depressive symptoms	Reboxetine
Gasser (2014)	In patients with anxiety associated with life-threatening diseases, reports of transient anxiety were 23% in the 200 µg LSD group and 50% in the placebo group	Phase II clinical trial	12	Transient anxiety symptoms	Unspecified benzodiazepines
Holze et al. (2021)	In healthy subjects, approximately 20% reported anxiety symptoms following a 200 µg dose of LSD	Phase I clinical trial	16	Transient anxiety symptoms	n/a
Kopra et al. (2022)	1.0% of LSD users reported seeking EMT, with a per-event risk estimate of 0.2%. Younger age, comorbid mental health conditions and higher frequency of use were associated with increased risk of EMT seeking. The most common symptoms were anxiety and panic	Observational study	10,293	Anxiety	n/a
Holze et al. (2022a)	In healthy subjects, 18% and 19% reported symptoms of transient anxiety following 100 µg and 200 µg of LSD, respectively	Phase I clinical trial	83	Transient anxiety symptoms	n/a
Holze et al. (2023)	Mild, transient anxiety and depression symptoms were reported by two and three patients, respectively, following 200 µg LSD. More severe albeit transient anxiety and delusions were reported by one patient during the LSD session, requiring rescue medication	Phase II clinical trial	42	Transient anxiety or depression symptoms	Lorazepam and olanzapine

Abbreviations: EMT: Emergency medical technician; HPPD: Hallucinogen-persisting perceptual disorder; LSD: Lysergic acid; MDMA: 3,4-Methylenedioxymethamphetamine

Table 3 Summary of studies examining N-N-DMT and ayahuasca-related symptoms of anxiety and affective disorders

Study	Major findings	Study design	N	Disorder	Treatment
Strassman et al. (1994)	In a cohort of healthy participants, all subjects who received the high-dose N-N-DMT (0.4 mg/kg) intravenously reported transient anxiety related to the intensity and speed of onset of administration. This anxiety alternated or coexisted with euphoria	Phase I clinical trial	12	Transient anxiety symptoms	n/a
Riba et al. (2001)	Among healthy volunteers, one participant experienced an intensely dysphoric reaction with transient anxiety at the medium dose of ayahuasca (0.75 mg N-N-DMT/kg)	Phase I clinical trial	6	Transient anxiety symptoms	n/a
Paterson et al. (2015)	A 42yo male presented with psychosis and anxiety following recent N-N-DMT use and long-term cannabis use	Case report	1	Psychosis and anxiety	Quetiapine, divalproex sodium and gabapentin
Brown et al. (2017)	A 40yo retired male psychiatrist with pre-existing bipolar I disorder presented with mania and psychosis following an attempt to self-medicate bipolar depression with N-N-DMT and phenelzine	Case report	1	Psychosis and mania	Lithium, clonazepam and paliperidone
D'Souza et al. (2022)	In a study of participants with treatment-resistant depression and healthy controls, six reported transient anxiety at drug onset (0.1–0.3 mg/kg N-N-DMT)	Phase I clinical trial	12	Transient anxiety symptoms	n/a
Good et al. (2023)	Among healthy volunteers, one participant reported transient anxiety symptoms following N-N-DMT IV infusion (9–21.5 mg)	Phase I clinical trial	24	Transient anxiety symptoms	n/a
Vogt et al. (2023)	N-N-DMT bolus (25 mg) + infusion (1.0 mg/min) doses produced higher ratings of anxiety than the IV infusion alone (1.0 mg/min) (5D-ASC)	Phase I clinical trial	27	Transient anxiety symptoms	n/a
Luan et al. (2024)	Among healthy volunteers, one participant reported transient anxiety symptoms following N-N-DMT administration (14–56.5 mg)	Phase I clinical trial	11	Transient anxiety symptoms	n/a

Abbreviations: DMT: Dimethyltryptamine

3.2.2 LSD

Nine studies showing LSD-related anxiety and affective symptoms were identified in the search (Table 2). An early case study reported symptoms of anxiety and panic following LSD ingestion in a college student which was resolved with psychotherapy (Boer and Sipprelle 1969). The author noted that the patient had a stressful relationship with his father and suggested that the anxiety symptoms did not originate from LSD use but were instead triggered by family separation. Anxiety symptoms following LSD use were also reported in a case series whereby 5 patients (aged 19–27) were admitted for LSD overdose (Blaho et al. 1997). All patients had attended a concert and reported co-ingestion of alcohol. Lorazepam was used to treat panic reactions and acute anxiety (Blaho et al. 1997). Another case report documented symptoms of mania following LSD use (Lake et al. 1981). The report detailed a 31-year-old male who had no personal or family history of psychiatric disorders but had been consuming alcohol heavily due to his recent separation from his wife. He developed symptoms of mania 2 weeks after LSD use which was successfully cleared with 3 weeks of lithium treatment (Lake et al. 1981). The emergence of affective symptoms related to LSD ingestion was reported in the case of a 26-year-old male (Lerner et al. 2002). He had a history of cannabis, MDMA and LSD use and reported perceptual disturbances lasting longer than expected and feeling like he was losing control. He suspended LSD and MDMA use but continued using cannabis. Two months later he suddenly developed hallucinogen-persisting perception disorder (HPPD) and experienced an MDD episode which was treated with reboxetine (Lerner et al. 2002). Furthermore, an observational study assessed LSD-related adverse experiences resulting in emergency medical treatment (EMT) seeking. The study found that 1.0% of LSD users reported seeking EMT and the most common symptoms were anxiety and panic ($n = 10,293$; Kopra et al. 2022). Younger age, comorbid mental health conditions and higher frequency of use were associated with increased risk of EMT seeking (Kopra et al. 2022). In all clinical trials that used LSD in the systematic review, only transient anxiety symptoms were reported (Gasser et al. 2014; Holze et al. 2021, 2022a, 2023). Two cases warranted the use of rescue medications to treat acute anxiety and/or delusions (Gasser et al. 2014; Holze et al. 2023).

3.2.3 DMT and Ayahuasca

The search identified six studies showing pure DMT (N-N-DMT) and two studies showing ayahuasca-related symptoms of anxiety and affective disorders (Table 3). A case report documented a 42-year-old male who presented with psychosis and anxiety symptoms following recent and repeated N-N-DMT use (Paterson et al. 2015). Notably, the patient had a family history of alcoholism, BP and OCD as well as a personal history of substance-use disorder (SUD) with long-term cannabis use. The patient also had a variety of psychosocial stressors including recent

unemployment, eviction from his apartment and the death of his mother. Treatment was successful with quetiapine for psychosis, divalproex sodium for impulsivity and gabapentin for anxiety (Paterson et al. 2015). Another study reported the case of a 40-year-old retired male psychiatrist who attempted to treat his pre-existing bipolar I disorder with N-N-DMT. The patient had one previous episode of mania but was otherwise depressed for most of his life. The patient reported taking 1 g vaporized N-N-DMT daily for the past 6 months and added 60 mg phenelzine daily 3 weeks before hospitalization. The patient was treated with lithium and clonazepam to target mania and paliperidone for psychosis (Paterson et al. 2015). In all clinical trials that used N-N-DMT (Strassman et al. 1994; D'Souza et al. 2022; Good et al. 2023) or ayahuasca (Riba et al. 2001; Luan et al. 2024) in the systematic review, only transient anxiety symptoms were reported. There were no rescue medications required or symptoms that persisted long-term. Furthermore, one trial with a cross-over design showed that the patients reported higher ratings of anxiety when N-N-DMT was administered via a bolus dose + infusion compared to an infusion alone (Vogt et al. 2023).

3.2.4 Combination of Classic Psychedelics

Five studies showing anxiety and affective symptoms related to the use of multiple psychedelic drugs were identified in the search (Table 4). One case reported documented the emergence of PTSD symptoms following concurrent LSD and N-N-DMT use (Rubin-Kahana et al. 2021). Specifically, a 25-year-old man recovered a repressed childhood memory of sexual abuse during the psychedelic experience. The patient had no psychiatric history but reported that both patients had opioid use disorder and that he used cannabis daily, as well as opioids, cocaine and LSD on a regular basis. The patient increased opioid use following the psychedelic experience to manage his PTSD symptoms. Methadone and sertraline were prescribed to manage opioid use and PTSD symptoms, respectively. Another study examined the association between lifetime hallucinogen use, abuse and dependence and panic attacks. Specifically, psychedelic use was defined as the use of psychedelics on at least five occasions. Conversely, psychedelic abuse was defined as the use of psychedelics causing substantial problems with family, friends, work or school and/or mental or physical health problems. Psychedelic dependence was defined as three or more of the following symptoms resulting from psychedelic use: (a) tolerance, (b) withdrawal symptoms, (c) unsuccessful effort to reduce or control use, (d) considerable problems with family, friends, work or school, (e) mental or physical health problems. The authors found that hallucinogen abuse and dependence, but not use, were significantly related to an increased lifetime risk of panic attacks (Bonn-Miller et al. 2007). Moreover, data from the national survey on drug use and health in the United States assessed past-year hallucinogen use in relation to psychological distress, depression and suicidality in adults (Yang et al. 2022). Findings from 241,675 respondents showed that LSD and N-N-DMT use was associated with an increased likelihood of suicidal planning. In addition, LSD use

Table 4 Summary of studies examining symptoms of anxiety and affective disorders related to the use of multiple psychedelics

Paper	Major findings	Study design	N	Disorder	Drug	Treatment
Bonn-Miller et al. (2007)	After controlling for relevant variables, hallucinogen abuse and dependence, but not use, were significantly related to an increased lifetime risk of panic attacks	Retrospective online survey	4,745	Panic disorder	LSD, mescaline, peyote, psilocybin, N-N-DMT	n/a
Yang et al. (2022)	LSD use was associated with an increased likelihood of MDE and suicidal thinking. N-N-DMT was associated with an increased likelihood of suicidal planning	Retrospective online survey	241,675	Depression	LSD, N-N-DMT	n/a
Rubin-Kahana et al. (2021)	A 25yo man developed PTSD following the use of LSD and N-N-DMT. In the psychedelic experience, a repressed memory of childhood sexual abuse was recovered	Case report	1	PTSD	LSD, N-N-DMT	Methadone and sertraline
Holze et al. (2022b)	In healthy subjects, low mood was reported in two subjects following psilocybin (15–30 mg) and two subjects following LSD (100–200 µg)	Phase I clinical trial	28	Transient affective symptoms	LSD, psilocybin	n/a
Simonsson et al. (2024)	In a cohort of adolescent twins, psychedelic use was significantly associated with more manic symptoms for individuals with a higher genetic vulnerability to schizophrenia or bipolar I disorder	Cross-sectional study	16,255	Manic symptoms	LSD, psilocybin	n/a

Abbreviations: DMT: Dimethyltryptamine; LSD: Lysergic acid; MDE: Major depressive episode; PTSD: Post-traumatic stress disorder

was also associated with an increased likelihood of MDD episodes (Yang et al. 2022). A recent cross-sectional study in a cohort of adolescent twins found that LSD or psilocybin use was significantly associated with more manic symptoms for individuals with a higher genetic vulnerability to schizophrenia or bipolar I disorder (Simonsson et al. 2024). A phase I clinical with a crossover design assessed the acute effects of these drugs in healthy subjects. Regarding adverse effects, low mood was reported in 2 subjects following psilocybin and 2 subjects following LSD (Holze et al. 2022b).

4 Discussion

We conducted a systematic review of existing literature concerning psychedelic-related anxiety and affective symptoms, identifying 39 studies in total. Specifically, there were associations between the use of classic psychedelics and the emergence of anxiety, panic disorder, PTSD, MDD and mania symptoms and occasionally, disorders. Anecdotal observations have also reported the development of OCD symptoms following psychedelics. Most clinical trials reported only transient anxiety symptoms following the ingestion of psychedelics, typically emerging and resolving during the dosing session with no requirement for rescue medications or additional psychological support beyond the integration sessions. Adverse event reporting in clinical trials was complicated by samples comprising participants with pre-existing psychiatric disorders. This made it challenging in some cases (e.g. individuals with anxiety or MDD) to distinguish between symptom induction/worsening of symptoms and non-responsiveness to psychedelics as a therapeutic intervention. In addition, in many clinical trials with psychedelics, adverse events may be underreported due to a lack of systematic assessment. While instances of psychedelics clearly triggering anxiety or affective disorders were rare, they were documented in a few of the studies. In almost all of these, the afflicted individuals presented with various risk factors, including polydrug use, overdose, personal and familial psychiatric histories, current psychosocial stressors and/or psychedelic use in a poor setting. These risk factors are closely intertwined with the importance of ‘set and setting’, a phenomenon underscoring how an individual’s pre-existing mental state (set) along with the physical and social environment in which the drug is ingested (setting) strongly influences the nature of psychedelic experience. These findings also highlight the importance of thorough screening in clinical trials and ensuring appropriate set and setting before psychedelic use. It is important to note that even despite these measures, instances of new or exacerbated psychiatric symptoms may occur in clinical trial settings. For example, adverse effects including suicidal ideation (Goodwin et al. 2022) and depressive symptoms (Goodwin et al. 2023; Schneier et al. 2023; Sloshower et al. 2023) have been reported in clinical trials using psychedelics.

4.1 Risk Factors for Developing Psychedelic-Related Anxiety and Affective Symptoms

This systematic review identified various risk factors for psychedelic-related anxiety and affective disorders. A personal psychiatric history was frequently noted as a risk factor across different psychedelics. Specifically, psychedelic-related mania was reported in several studies with pre-existing psychiatric illnesses including MDD and PTSD (Hendin and Penn 2021), MDD and GAD (Barber et al. 2022) as well as bipolar disorder (Brown et al. 2017). Substance-use disorders were also associated with instances of psychedelic-related psychosis and anxiety (Paterson et al. 2015) and PTSD (Rubin-Kahana et al. 2021). Additionally, a family history of bipolar disorder has been cited in cases of psilocybin-related mania (Hendin and Penn 2021; Barber et al. 2022) and DMT-related psychosis and anxiety (Paterson et al. 2015). The review also highlighted the importance of considering trauma history. For instance, one case study outlined the emergence of PTSD symptoms after a repressed memory of childhood sexual abuse surfaced during a psychedelic experience (Rubin-Kahana et al. 2021). Whilst this doesn't exclude individuals with a trauma history from participating in psychedelic trials, it underscores the necessity of supervised psychedelic use and professional psychological support before, during and after the psychedelic experience.

Polydrug use also emerged as a significant risk factor. Reports include anxiety symptoms following LSD use and co-ingestion of alcohol (Blaho et al. 1997) and psychosis and mania from daily N-N-DMT and phenelzine used to self-medicate bipolar depression (Brown et al. 2017). Moreover, frequent marijuana use was a common denominator in several cases (Benjamin 1979; Lerner et al. 2002; Paterson et al. 2015; Rubin-Kahana et al. 2021; Suleiman et al. 2022). This was associated specifically with the development of panic disorder (Benjamin 1979), mania (Suleiman et al. 2022), HPPD and depressive symptoms (Lerner et al. 2002), psychosis and anxiety (Paterson et al. 2015) as well as psychosis and mania (Brown et al. 2017). Dosing and administration practices remain important considerations, with reported cases of anxiety symptoms (Blaho et al. 1997) and panic disorder (Benjamin 1979) linked to overdoses. Moreover, a clinical study highlighted differential levels of anxiety attributed to the route of administration, with bolus doses + infusion producing higher ratings of anxiety compared to IV infusion alone (Vogt et al. 2023). Previous psychedelic use does not seem to be a risk factor as studies in the systematic review showed the emergence of psychedelic-related anxiety and affective symptoms in those who had previously used psychedelics (Boer and Sipprelle 1969; Benjamin 1979; Blaho et al. 1997) as well as psychedelic naïve individuals (Hendin and Penn 2021). It is important to note that increased lifetime use does not reduce the risk of psychedelic-related anxiety and affective symptoms. Instead, it is likely that those who suffer from such symptoms following initial psychedelic use are not likely to continue to use psychedelics.

The presence of current psychosocial stressors was another well-established risk factor, with cases reporting that individuals experiencing major life stressors such as

unemployment, the death of a loved one (Paterson et al. 2015) or stressful family relationships (Boer and Sippelle 1969; Benjamin 1979; Lake et al. 1981) were more likely to develop anxiety and affective disorders following psychedelic use. Such stressors can substantially destabilize the psychological state of individuals and consequently predispose the development of anxiety and affective disorders following psychedelic use. Moreover, the setting in which psychedelics are consumed heavily influences the emergence of adverse effects. Almost all reported psychedelic-related anxiety and affective symptoms arose in recreational settings lacking structured psychological preparation and support. This was underscored by one study that documented five cases of anxiety in patients who overdosed on LSD at a concert. This is in stark contrast to controlled, therapeutic settings that implement extensive safety measures to reduce the risk of harm, including thorough screening, patient monitoring, supervised dosing sessions in a structured environment as well as psychological preparation and integration. This again underscores the importance of set and setting, emphasizing the need for individuals to enter the dosing session with a stable mental state within a safe and comfortable physical and social environment.

4.2 *Acute Transient Anxiety*

Whilst examining risk factors and rescue medications is crucial in addressing severe cases of anxiety and affective disorders, it is also important to discuss the nature of acute transient anxiety that can arise during psychedelic experiences. Challenging experiences are a well-established phenomenon in psychedelic dosing sessions and tools such as the Challenging Experiences Questionnaire (CEQ; Barrett et al. 2016) are frequently employed in clinical studies to understand the psychologically difficult aspects of psychedelic experiences. Challenging experiences are not solely pathological and can often be understood as part of the therapeutic process, particularly when they coincide with emotional breakthroughs (Haijen et al. 2018). Carbonaro et al. (2016) found that the duration of the difficulty associated with the challenging experience negatively correlated with improved well-being. Thus, it is important to distinguish between challenging experiences which yield adverse effects or no therapeutic changes and those which may be cathartic and lead to therapeutic benefit. For more information on managing such experiences in psychedelic-assisted therapy, see Chapter: Guiding Through Challenging Experiences and ‘Bad Trips’ (Gasser and Koslowski).

In most cases reviewed here, anxiety emerged as a transient symptom during the dosing session rather than manifesting as a sustained clinical condition. Such transient anxiety may be a normal and expected part of the therapeutic process, similar to prolonged exposure therapy, one of the most effective treatments for PTSD (Brown et al. 2019). This therapy involves increased increments of exposure to trauma-related content and encourages patients to process distressing memories until their distress diminishes or becomes manageable. Moreover, anxiety may also

play a functional role in PAT. Neuroimaging studies have identified a neurobiological correlate of anxiety during the acute effects of psychedelics. Whilst selective serotonin reuptake inhibitors (SSRIs) typically reduce amygdala responses to emotional stimuli (Ma 2015), a clinical study on patients with treatment-resistant depression showed that psilocybin had the opposite effect (Roseman et al. 2019). In this study, amygdala activity was recorded in response to the presentation of neutral, fearful and happy faces in an fMRI scanner before and after a dosing session. The findings revealed increased responses to both fearful and happy faces in the right amygdala post-treatment and that this enhanced activation was linked to clinical improvement (Roseman et al. 2019). The contrasting effects of SSRIs and psilocybin were attributed to different therapeutic actions; SSRIs tend to mitigate or blunt negative emotion whereas psilocybin encourages patients to confront and process negative emotions. This emotional confrontation can also be applied to a cognitive-behavioural framework of PAT's psychological mechanisms (Wolff et al. 2020). Under this framework, psychopathology is rooted in experiential avoidance, that is, the attempt to evade escape or alter private events (e.g. emotions, thoughts, memories, etc.) despite harmful consequences. The ability to allow private events to unfold without attempting to control them reduces psychopathology and is termed acceptance. Wolff et al. (2020) propose that psychedelics relax avoidance-related beliefs in a way that can facilitate acceptance. Within this context, challenging experiences during PAT are viewed as a form of experiential avoidance, where the individual attempts to resist emotional confrontation. This also marks the beginning of an acceptance-promoting learning process. The authors suggest that there may be therapeutic value in repeated dosing sessions whereby challenging experience lessens and the individual may revise avoidance-related beliefs.

4.3 *Treatment*

In some severe cases, medication was employed to address psychedelic-related anxiety and affective disorders. Benzodiazepines, particularly lorazepam, were commonly prescribed to treat anxiety symptoms across several studies (Benjamin 1979; Blaho et al. 1997; Gasser et al. 2014; Brown et al. 2017; Goodwin et al. 2022; Holze et al. 2023). For symptoms of mania and psychosis, a regimen typically included second-generation atypical antipsychotics such as olanzapine (Suleiman et al. 2022; Holze et al. 2023), quetiapine (Paterson et al. 2015) and paliperidone (Brown et al. 2017). Lithium was also used to treat mania in three cases (Lake et al. 1981; Brown et al. 2017; Suleiman et al. 2022) and divalproex sodium in two instances (Paterson et al. 2015; Suleiman et al. 2022).

4.4 Limitations

This systematic review is subject to methodological limitations that must be addressed to draw more robust conclusions. A primary limitation is the heterogeneity in study designs. The review comprised phase I and II clinical trials, observational studies, longitudinal studies, retrospective surveys, cross-sectional studies as well as case reports. This diversity complicates the ability to generalize findings due to varying methodologies, sample sizes and research contexts. The lack of controlled environments in many of these studies presents significant challenges in distinguishing between correlation and causation, as these studies typically do not control for confounding variables, nor do they provide a basis for establishing a causal relationship between psychedelic use and psychological outcomes. The predominance of online surveys and case reports further undermines the ability to draw definitive conclusions about the temporal relationship between psychedelic use and anxiety and affective disorders. To overcome these limitations, there is a critical need for systematic research that incorporates multi-method designs, including both qualitative and quantitative approaches as well as longitudinal studies. Such designs would help delineate the nuances between transient and sustained psychedelic-related anxiety and affective symptoms and clarify the causal pathways involved in therapeutic outcomes versus adverse effects. Moreover, the current assessment and reporting standards of adverse events in PAT trials lack standardization and specificity to the features of PAT (Palitsky et al. 2024). A prime example of this issue is the reporting of anxiety. In this context, the experience of transient anxiety arising from the acute drug effects which may lead to therapeutic improvement does not share the same phenomenological features as anxiety recognized as a clinical diagnosis. Thus, in this instance, AE reporting guidelines are poorly calibrated to the features of PAT. These guidelines could be improved with the addition of a specifier which clarifies the duration of this adverse event and whether it was associated with clinical improvement, thereby providing clearer and more useful information.

4.5 Implications

The findings of the systematic review have important implications for the use of medication in treating both acute symptoms and long-term disorders. The case studies highlight the critical importance of proper set, setting and screening. Most of these patients would likely have not been considered suitable candidates for PAT. A combination of personal or familial psychiatric history along with current psychosocial stress could lead to a mental state that is too unstable to safely experience drug-induced alterations in consciousness. Moreover, this precarious psychological baseline when combined with unsafe dosing practices (overdoses, polydrug use) or an inappropriate setting significantly heightens the risk of adverse reactions. Clinical

trials employing PAT adhere to strict guidelines and structures to maximize patient safety. This process begins with thorough screening to prevent high-risk patients from participating. Common exclusion criteria in these studies often include a personal or family history of psychosis, schizophrenia or bipolar; clinically significant suicidal ideation; failure to establish rapport with a therapist during preparation sessions; inability/unwillingness to down-titrate medication or a positive urine test for illicit drugs.

The formal phases of psychological preparation, supervised dosing and integration therapy are also constructed in a way that safeguards patient well-being. However, additional measures can be taken to further mitigate the risk of psychedelic-related anxiety and affective symptoms. A recent qualitative study explored patient experiences with psychedelic treatments and created a list of recommendations to optimize treatment approaches (Breeksema et al. 2024). During the preparatory phase, establishing a strong rapport between the patient and therapist is crucial as a robust therapeutic alliance has been associated with improved therapeutic response in multiple studies (Murphy et al. 2022; Levin et al. 2024). Allocating sufficient time for preparation sessions allows trust to develop, which is especially important as some patients may require more time than others to feel comfortable. Participants may also benefit from practical instructions on navigating and surrendering to the experience and any challenging emotions that arise during the session. Additionally, involving former patients in peer-to-peer education can help normalize and validate the experiences of those undergoing psychedelic-assisted therapy (Breeksema et al. 2024).

Regarding the dosing session, it may be beneficial for participants to gradually build up to the high dose to allow participants to acclimate to the altered state of consciousness and thus prevent associated anxiety. This may be especially important for individuals with no prior experience with psychedelics. The use of music, tailored to individual preferences, can also provide comfort during the session. Because music is strongly tied to personal preference, a personally curated playlist with familiar music can enhance the therapeutic experience by facilitating a sense of surrender (Breeksema et al. 2024). In the integration phase, the type and frequency of integration therapy sessions should be customized based on patient characteristics and circumstances. Moreover, the integration of psychedelic experience may require more time than study contexts accommodate. While ongoing therapeutic support is ideal, it might not always be feasible. In such cases, community support groups with other patients may partially bridge this gap (Breeksema et al. 2024). Collectively, these recommendations can optimize treatment approaches and ensure patient safety is upheld to the highest standard.

4.6 Conclusion

To the best of our knowledge to date, this is the first systematic review to synthesize the current literature assessing psychedelic-related anxiety and affective disorders. In

most clinical trials, only transient anxiety symptoms were present following the ingestion of psychedelics and these symptoms typically emerged and dissipated during the dosing session with no requirement for rescue medications. Such cases warrant a consideration of nuance when it comes to the categorization of this anxiety as an adverse effect. Cases where psychedelics led to clinically significant anxiety or affective symptoms were infrequent but mostly occurred in patients who presented with a combination of risk factors and ingested psychedelics in recreational settings. Such risk factors included a personal or family history of psychiatric disorders, current psychosocial stress, polydrug use, interaction with medication and overdose. However, there remains a realistic risk of psychedelics leading to aggravations of pre-existing disorders or the emergence of new psychiatric symptoms. It has not yet been understood which patients will have challenging experiences, bad trips or poor therapeutic outcomes. Only when the risks and dangers of psychedelics are properly assessed and understood by investigating them can a reasonable risk–benefit analysis be reached.

Appendices

Appendix 1: PRISMA Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Title			
Title	1	Identify the report as a systematic review	1
Abstract			
Abstract	2	See the PRISMA 2020 for Abstracts checklist	1
Introduction			
Rationale	3	Describe the rationale for the review in the context of existing knowledge	2–3
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses	2–3
Methods			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses	4
Information sources	6	Specify all databases, registers, websites, organizations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted	4
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used	4

(continued)

Section and Topic	Item #	Checklist item	Location where item is reported
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process	4
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process	4
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect	4
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information	4
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process	4
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results	4
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5))	5
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions	n/a
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses	5
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the	n/a

(continued)

Section and Topic	Item #	Checklist item	Location where item is reported
		model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used	
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression)	n/a
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results	n/a
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases)	4
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome	4
Results			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram	5
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded	5
Study characteristics	17	Cite each included study and present its characteristics	6–13
Risk of bias in studies	18	Present assessments of risk of bias for each included study	26–29
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots	6–13
Results of syntheses	20a	For each synthesis, briefly summarize the characteristics and risk of bias among contributing studies	6–13; 26–29
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect	n/a
	20c	Present results of all investigations of possible causes of heterogeneity among study results	n/a
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results	n/a

(continued)

Section and Topic	Item #	Checklist item	Location where item is reported
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed	26–29
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed	n/a
Discussion			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence	14
	23b	Discuss any limitations of the evidence included in the review	17–18
	23c	Discuss any limitations of the review processes used	17–18
	23d	Discuss implications of the results for practice, policy and future research	18–19
Other information			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered	4
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared	4
	24c	Describe and explain any amendments to information provided at registration or in the protocol	n/a
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review	n/a
Competing interests	26	Declare any competing interests of review authors	n/a
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: Template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review	n/a

Appendix 2: Quality Assessment (Tables 5, 6, 7, and 8)

Table 5 Quality assessment of case reports using JBI critical appraisal checklist for case reports (Moola et al. 2020)

	Boer and Sippelle (1969)	Benjamin (1979)	Lake et al. (1981)	Blaho et al. (1997)	Lerner et al. (2002)	Paterson et al. (2015)	Brown et al. (2017)	Hendin and Penn (2021)	Rubin-Kahana et al. (2021)	Barber et al. (2022)	Suleiman et al. (2022)
Were patient's demographic characteristics clearly described?	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Was the patient's history clearly described and presented as a timeline?	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y
Was the current clinical condition of the patient on presentation clearly described?	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y
Were diagnostic tests or assessment methods and the results clearly described?	N	N	N	Y	Y	U	Y	N	N	Y	Y
Was the intervention(s) or treatment procedure(s) clearly described?	Y	Y	U	Y	Y	Y	Y	U	Y	Y	Y
Was the post-intervention clinical condition clearly described?	Y	N	U	Y	Y	Y	N	U	U	Y	U
Were adverse events (harms) or unanticipated events identified and described?	U	N	Y	Y	Y	Y	Y	Y	Y	Y	Y
Does the case report provide takeaway lessons?	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y

Y = yes; N = no; U = unclear

Table 6 Quality assessment of randomized controlled trials using the Cochrane risk of bias tool (Sterne et al. 2019)

Study	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other biases
Studerus et al. (2011)	Low	Low	Low	Low	Low	Low	–
Gasser et al. (2014)	Low	Low	Low	Low	Low	Low	–
Griffiths et al. (2016)	Low	Low	Low	Low	Low	Low	–
Davis et al. (2021)	Low	Low	Low	Low	Low	Low	–
Goodwin et al. (2022)	Low	Low	Low	Low	Low	Low	–
Goodwin et al. (2023)	Low	Low	Low	Low	Low	Low	–
Holze et al. (2023)	Low	Low	Unclear	Low	Low	Low	–

Table 7 Quality assessment of open-label studies using the modified Newcastle-Ottawa scale (Deeks et al. 2003)

Study	Selection		Outcomes			Total (/5)
	Representativeness	Ascertainment of exposure	Assessment	Follow-up	Adequacy of follow-up	
Strassman et al. (1994)		*	*			**
Riba et al. (2001)		*	*			**
Hasler et al. (2004)	*	*	*			***
Griffiths et al. (2006)	*	*	*	*	*	*****
Griffiths et al. (2011)	*	*	*	*	*	*****
Carhart-Harris et al. (2016)	*	*	*	*	*	*****
Holze et al. (2021)	*	*	*			***
D'Souza et al. (2022)	*	*	*			***
Holze et al. (2022a)	*	*	*			***
Holze et al. (2022b)	*	*	*			***
Rucker et al. (2022)	*	*	*	*	*	*****
Good et al. (2023)		*	*			**
Schneier et al. (2023)	*	*	*	*	*	*****
Sloshower et al. (2023)	*	*	*	*	*	*****
Vogt et al. (2023)	*	*	*			***
Luan et al. (2024)	*	*	*			***

* Poor quality; **_ Fair quality; ***_ Good quality

Table 8 Quality assessment of observational studies using the CASP cohort studies checklist (CASP 2018)

	Bonn-Miller et al. (2007)	Kopra et al. (2022)	Yang et al. (2022)	Nayak et al. (2023)	Simonsson et al. (2024)
Did the study address a clearly focused issue?	Y	Y	Y	Y	S
Was the cohort recruited in an acceptable way?	Y	Y	Y	Y	S
Was the exposure accurately measured to minimize bias?	N	N	N	N	N
Was the outcome accurately measured to minimize bias?	S	Y	S	S	Y
(a) Have the authors identified all confounding factors?	N	N	N	N	N
(b) Have they taken account of the confounding factors in the design and/or analysis?	N	N	S	N	S
(c) Was the follow-up of subjects complete enough?	N/A	N/A	N/A	Y	Y
(d) Was the follow-up of subjects long enough?	N/A	N/A	N/A	Y	Y
Are the results precise?	Y	Y	S	Y	Y
Do you believe the results?	Y	Y	S	Y	Y
Can the results be applied to the local population?	Y	Y	S	Y	Y
Do the results of this study fit with other available evidence?	Y	Y	S	Y	Y
Does the study have implications for practice?	Y	Y	Y	Y	Y
Total (/26)	15	16	12	19	19

Y = Yes; N = No, S = Somewhat; N/A = Not available

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